

EC-467 Mobile Application Development
BS (CE)-2021
Project Report of Prayer Alarm



Submitted By:

Reg. No.	Name
21-CE-009	Huzaifa Shahbaz

Submitted To:

Sir Muhammad Bilal

Department of Computer Engineering
HITEC University Taxila

Prayer Alarm

Contents

CHAPTER#1 ABSTRACT	6
1.1 Abstract:	6
CHAPTER#2 INTRODUCTION	6
2.1 Introduction:	6
CHAPTER#3 OBJECTIVE	6
3.1 Objective:	6
CHAPTER#4 PROBLEM STATEMENT	7
4.1 PROBLEM STATEMENT:	7
CHAPTER#5 EXPECTED OUTCOME	7
5.1 EXPECTED OUTCOME:	7
CHAPTER#6 COMPONENTS USED	7
6.1 COMPONENTS USED:	7
CHAPTER#7 ANDROID STUDIO JELLYFISH	8
7.1 ANDROID STUDIO JELLYFISH:	8
CHAPTER#8 AVD EMULATOR	8
8.1 AVD EMULATOR:	8
CHAPTER#9 JAVA	8
9.1 JAVA:	8
CHAPTER#10 XML IN LINEAR LAYOUT	9
10.1 XML IN LINEAR LAYOUT:	9
CHAPTER#11 FUNCTIONAL REQUIREMENTS	9
11.1 FUNCTIONAL REQUIREMENTS:	9
11.1.1 Alarm Setting:	9
11.1.2 CUSTOMIZATION:	9
11.1.3 REPEAT OPTIONS:	9
11.1.4 SNOOZE FUNCTIONALITY:	9
11.1.5 SILENT MODE HANDLING:	10
11.1.6 AUTOMATIC TIMEZONE ADJUSTMENT:	10
11.1.7 NOTIFICATION HANDLING:	10
11.1.8 ACCESSIBILITY FEATURES:	10

11.1.9 INTEGRATION WITH CALENDAR:	10
11.1.10 BACKUP AND RESTORE:	10
11.1.11 MULTIPLE LANGUAGE SUPPORT:	10
11.1.12 OFFLINE FUNCTIONALITY:	10
CHAPTER#12 NON-FUNCTIONAL REQUIREMENTS	10
12.1 NON-FUNCTIONAL REQUIREMENTS:	10
12.1.1 PERFORMANCE:	10
12.1.2 REALIABILITY:	11
12.1.3 USABILITY:	11
12.1.4 ACCESSIBILITY:	11
12.1.5 Security:	11
12.1.6 Compatibility:	11
12.1.7 Maintainability:	11
12.1.8 Availability:	11
12.1.9 Efficiency:	11
12.1.10 Localization:	11
Chapter#13 Main Activity of Prayer Alarm and Code	11
13.1 Main Activity of Prayer Alarm:	11
13.2 JAVA CODE OF MAIN ACTIVITY:	12
13.3 XML CODE OF PRAYER ALARM:	14
13.4 JAVA CODE OF ALARM RECEIVER:	14
13.5 OUTPUT OF MAIN ACTIVITY:	16
CHAPTER#14 Multiple Activities and Intent AND CODES	16
14.1 Multiple Activities and Intent:	16
14.2 JAVA CODE OF MAINACTIVITYINTENT:	17
14.3 XML CODE OF INTENT:	18
14.4 JAVA CODE OF ALARM RECEIVER:	19
14.5 OUTPUT OF INTENT:	20
CHAPTER#15 LIST VIEW AND ARRAY ADAPTER AND CODE	20
15.1 LIST VIEW AND ARRAY ADAPTER:	20
15.2 JAVA CODE OF LIST VIEW AND ARRAY ADAPTER:	21
15.3 XML CODE OF ARRAYADAPTERANDLISTVIEW:	22
15.4 JAVA CODE OF BROADCASTRECEIVER:	23

15.5 JAVA CODE OF ALARM RECEIVER:	24
15.6 OUTPUT OF LIST VIEW AND ARRAY ADAPTER:	25
CHAPTER#16 LIST VIEW AND BASE ADAPTER AND CODES	25
16.1 LIST VIEW AND BASE ADAPTER:	25
16.2 JAVA CODE OF BASEADAPTERANDLISTVIEW:	26
16.3 XML CODE OF BASEADAPTERANDLISTVIEW:	27
16.4 JAVA CODE OF ALARM RECEIVER:	28
16.5 JAVA CODE OF PRAYERTIMEADAPTER:	29
16.6 XML CODE OF PRAYERTIMEADAPTER:	30
16.7 OUTPUT OF BASEADAPTERANDLISTVIEW:	31
CHAPTER#17 SHARED PREFERENCES	32
17.1 SHARED PREFERENCES:	32
17.2 JAVA CODE OF SHARED PREFERENCES:	32
17.3 XML CODE OF SHARED PREFERENCES:	34
17.4 JAVA CODE OF ALARM RECEIVER:	34
17.5 OUTPUT OF SHARED PREFERENCES:	36
CHAPTER#18 VIEW PAGER AND CODES	36
18.1 VIEW PAGER:	36
18.1.1 Welcome Screen:	36
18.1.2 Set Prayer Times:	37
18.1.3 Alarm Settings:	37
18.1.4 Qibla Direction:	37
18.1.5 Daily Prayer Tracker:	37
18.2 JAVA CODE OF VIEW PAGER:	37
18.3 XML CODE OF VIEW PAGER:	37
18.4 JAVA CODE OF PRAYER FRAGMENT:	38
18.5 XML CODE OF PRAYER FRAGMENT:	39
18.6 JAVA CODE OF VIEWPAGERADAPTER:	39
18.7 OUTPUT OF VIEW PAGER:	40
Chapter#19 Conclusion:	41
19.1 Conclusion:	41

Prayer Alarm

Figure 13.1:Main Activity of prayer alarm app	16
Figure 14.2:Intent activity of prayer alarm app	20
Figure 15.3:ARRAY ADAPTER AND LIST VIEW OF PRAYER ALARM APP	25
Figure 16.4:BASE ADAPTER AND LISTVIEW OF PRAYER ALARM.....	31
Figure 17.5:SHARED PERFERNCES OF PRAYER ALARM	36
Figure18.6:VIEW PAGER OF PRAYER ALARM APP	40

Prayer Alarm

CHAPTER#1 ABSTRACT

1.1 Abstract:

An app called Prayer Alarm is presented as a way for Muslim people to remember to pray every day even when they have a lot of work to do. By utilizing personalized alerts and inspirational elements, it seeks to promote spiritual awareness in addition to intellectual endeavors. The project aims to improve consumers' overall well-being while satisfying a technological requirement.

CHAPTER#2 INTRODUCTION

2.1 Introduction:

It can be difficult to recall daily prayers in our hectic academic lives. An approachable smartphone app to remind students of prayer times is provided by the Prayer Alarm initiative as a remedy. It encourages spiritual mindfulness in addition to academic endeavors with its motivational elements and individualized notifications. Our goal is to develop a product that enhances users' overall wellbeing while also meeting a technological necessity. Come along with us as we set out to balance spiritual commitment with academic excellence.

CHAPTER#3 OBJECTIVE

3.1 Objective:

The Prayer Alarm project aims to create a mobile application that serves as a reminder for students to pray every day, promoting spiritual attention in the midst of academic obligations. It strives to encourage regularity in prayer practice with features that are adjustable for notifications and motivation. The goal of the initiative is to give students a tool that fits in well with their daily routines and helps them better manage their time between spirituality and academic brilliance. Through tackling the difficulty of recalling prayers, it seeks to enhance users' overall health and spiritual satisfaction.

CHAPTER#4 PROBLEM STATEMENT

4.1 PROBLEM STATEMENT:

The challenge of students remembering daily prayers in the middle of busy academic schedules is the issue that the Prayer Alarm project attempts to solve. As a result, spiritual obligations are neglected and there is inconsistency. Reminder systems that are already in use frequently lack personalization and struggle to fit in with students' daily routines. As a result, a lot of students find it difficult to keep a healthy balance between their spiritual and intellectual endeavors. By offering a user-friendly solution that is customized to meet the needs of Muslim's people, The Prayer Alarm aims to close this gap and promote a harmonious integration of academic and spiritual life.

CHAPTER#5 EXPECTED OUTCOME

5.1 EXPECTED OUTCOME:

Muslim Students who participate in the Prayer Alarm initiative should be equipped to continue praying every day in a deliberate and regular manner. Muslim Students will find it easier to fit their spiritual responsibilities into their hectic academic schedules by using the application. Increased spiritual contentment, better time management abilities, and a better feeling of balance between one's personal and academic lives are expected outcomes of this. In the end, the Prayer Alarm hopes to support students' overall wellbeing by encouraging a closer relationship with their faith while they work toward their academic objectives.

CHAPTER#6 COMPONENTS USED

6.1 COMPONENTS USED:

The components used in prayer alarm system are:

- 1.Android studio Iguana
- 2.AVD Emulator
- 3.Java
- 4.Linear layout in XML

CHAPTER#7 ANDROID STUDIO JELLYFISH

7.1 ANDROID STUDIO JELLYFISH:

Android Studio is the main integrated development environment (IDE) for creating Android apps, and it is based on IntelliJ IDEA. It provides an extensive toolkit for creating, writing, testing, and debugging Android apps. It simplifies the app creation process with features like visual layout editor, code completion, and easy connection with the Android SDK. Developers are able to test programs on a variety of virtual devices thanks to its powerful emulator. Multiple programming languages, such as Java and Kotlin, are supported by Android Studio, enabling developers to construct inventive and feature-rich Android apps.

CHAPTER#8 AVD EMULATOR

8.1 AVD EMULATOR:

Without requiring actual devices, Android applications can be tested in a virtual environment with the help of the Android Virtual Device (AVD) emulator. Different device configurations, including screen size, resolution, and Android version, can be simulated by developers. To ensure compatibility and dependability, developers can test an alarm's performance across several virtual devices using the AVD emulator when creating a prayer alarm app. It improves the overall performance of the program by making debugging and fine-tuning its behavior under various circumstances easier. Because it offers a handy platform for iterative testing and improvement of the prayer alarm app, the AVD emulator speeds up the development process.

CHAPTER#9 JAVA

9.1 JAVA:

Java classes and methods can be used to construct a prayer alarm that manages event scheduling and triggers at certain periods. With classes like as 'Timer' or 'ScheduledExecutorService', programmers can configure recurring tasks to trigger the alert. User interactions and audio playback may require integration with platform-specific APIs. Furthermore, the integration of date and time utilities facilitates the accurate scheduling of alarms for prayer. By addressing any mistakes that might arise during the setup or operation of an alarm, exception handling guarantees dependability. Finally, when developing a Java-based prayer alarm

application, modular design and encapsulation facilitate code readability and maintainability.

CHAPTER#10 XML IN LINEAR LAYOUT

10.1 XML IN LINEAR LAYOUT:

The prayer alarm user interface (UI) can be organized in XML with a LinearLayout, featuring views for selecting the time, alarm setting toggles, and customization options. Within the linear layout, attributes such as `layout_width` and `layout_height` dictate the dimensions and orientation of views. Users can set alarm times and enable/disable alarms with widgets like TimePicker and Switch. Furthermore, Text-Views have the ability to name UI elements, assisting users in setting up alarms. Buttons can initiate actions such as saving alarm settings or closing the alarm configuration screen by using the `{onClick}` properties. Last but not least, setting `{orientation}` to "vertical" guarantees a layout that is stacked vertically, making it easier for users to interact with the prayer alarm interface.

CHAPTER#11 FUNCTIONAL REQUIREMENTS

11.1 FUNCTIONAL REQUIREMENTS:

Functional Requirements are the following are:

11.1.1 Alarm Setting:

Users should be able to set multiple alarms for specific prayer times.

11.1.2 CUSTOMIZATION:

The system should allow users to customize alarm tones, labels, and vibration settings.

11.1.3 REPEAT OPTIONS:

Users may need the option to repeat alarms daily, weekly, or on specific days.

11.1.4 SNOOZE FUNCTIONALITY:

Providing a snooze option allows users to temporarily silence alarms and be reminded again later.

11.1.5 SILENT MODE HANDLING:

The system should respect the device's silent mode preferences, allowing alarms to still sound when the device is muted.

11.1.6 AUTOMATIC TIMEZONE ADJUSTMENT:

Ensuring accurate prayer times by adjusting alarms based on the user's current time-zone.

11.1.7 NOTIFICATION HANDLING:

Displaying notifications before the alarm sounds, allowing users to dismiss or snooze alarms directly from the notification.

11.1.8 ACCESSIBILITY FEATURES:

Including features like large fonts, color contrast, and voice guidance for users with visual or auditory impairments.

11.1.9 INTEGRATION WITH CALENDAR:

Optionally, integrating with the device's calendar to set alarms based on pre-defined prayer schedules or events.

11.1.10 BACKUP AND RESTORE:

Supporting multiple languages to cater to a diverse user base.

11.1.11 MULTIPLE LANGUAGE SUPPORT:

Supporting multiple languages to cater to a diverse user base.

11.1.12 OFFLINE FUNCTIONALITY:

Allowing users to set and manage alarms even when the device is offline or without internet connectivity.

CHAPTER#12 NON-FUNCTIONAL REQUIREMENTS

12.1 NON-FUNCTIONAL REQUIREMENTS:

Non-functional requirements for a prayer alarm system are:

12.1.1 PERFORMANCE:

Ensuring that the app responds quickly to user interactions and that alarms are triggered promptly.

12.1.2 REALIABILITY:

Alarms should reliably sound at the specified times, without failures or delays.

12.1.3 USABILITY:

The user interface should be intuitive and easy to navigate, even for users with limited technical knowledge.

12.1.4 ACCESSIBILITY:

The app should be accessible to users with disabilities, following accessibility guidelines and providing features like screen readers and high contrast modes.

12.1.5 Security:

Ensuring that user data, including alarm settings and preferences, is securely stored and transmitted.

12.1.6 Compatibility:

The app should be compatible with various mobile devices and operating systems.

12.1.7 Maintainability:

The codebase should be well-documented and modular to facilitate updates and maintenance.

12.1.8 Availability:

The application should be available for use 24/7 with minimal downtime.

12.1.9 Efficiency:

The app should use device resources, such as battery and memory, efficiently to avoid draining the device.

12.1.10 Localization:

The app should support multiple languages and regional settings to cater to a diverse user base.

Chapter#13 Main Activity of Prayer Alarm and Code

13.1 Main Activity of Prayer Alarm:

The Prayer Alarm app accommodates a variety of religious customs by providing a customized reminder for prayer times. By providing their location and religious affiliation, users enable the app to generate precise prayer schedules. Depending on

the user's preferences, it notifies them at each prayer time. With features like silent mode for subtle reminders, the app makes sure that prayer routines are followed without interfering. Within the app, users can access additional resources including prayer guides and religious texts. By allowing users to interact with other believers and exchange prayer experiences, it promotes a sense of community. All in all, Prayer Alarm's smooth interface and extensive features encourage spiritual discipline and attentiveness.

13.2 JAVA CODE OF MAIN ACTIVITY:

```
package com.huzaifashahbaz.prayeralarmapp;
import android.app.AlarmManager;
import android.app.PendingIntent;
import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;
public class MainActivity extends AppCompatActivity {
    private EditText editTextTime;
    private Button buttonSetAlarm;
    private TextView textViewStatus;
    private AlarmManager alarmManager;
    private PendingIntent pendingIntent;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        editTextTime = findViewById(R.id.editTextTime);
        buttonSetAlarm = findViewById(R.id.buttonSetAlarm);
        textViewStatus = findViewById(R.id.textViewStatus);
        alarmManager=(AlarmManager)
getSystemService(Context.ALARM_SERVICE);
        buttonSetAlarm.setOnClickListener(v-> setAlarm());
    }
    private void setAlarm()
```

```

{
    String time=editTextTime.getText().toString();
    if (time.isEmpty())
    {
        Toast.makeText(this, "Please enter a time",
Toast.LENGTH_SHORT).show();
        return;
    }
    String[] timeParts=time.split(":");
    if (timeParts.length!=2)
    {
        Toast.makeText(this, "Invalid time format",
Toast.LENGTH_SHORT).show();
        return;
    }
    int hour,minute;
    try {
        hour=Integer.parseInt(timeParts[0]);
        minute=Integer.parseInt(timeParts[1]);
    }
    catch (NumberFormatException e)
    {
        Toast.makeText(this, "Invalid time format",
Toast.LENGTH_SHORT).show();
        return;
    }
    Calendar calendar=Calendar.getInstance();
    calendar.set(Calendar.HOUR_OF_DAY,hour);
    calendar.set(Calendar.MINUTE,minute);
    calendar.set(Calendar.SECOND,0);
    Intent intent=new Intent(this,AlarmReceiver.class);

    alarmManager.setExact(AlarmManager.RTC_WAKEUP,calendar.getTimeInMillis(
),pendingIntent);
    textViewStatus.setText("Alarm set for"+time);
    Toast.makeText(this, "Alarm set for", Toast.LENGTH_SHORT).show();
}
}

```

13.3 XML CODE OF PRAYER ALARM:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:tools="http://schemas.android.com/tools"
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <EditText
        android:id="@+id/editTextTime"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter prayer time(HH:MM)"
        android:inputType="time"
        android:padding="16dp"/>
    <Button
        android:id="@+id/buttonSetAlarm"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Set Alarm"
        android:padding="16dp"/>
    <TextView
        android:id="@+id/textViewStatus"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text=""
        android:padding="16dp"/>
</LinearLayout>
```

13.4 JAVA CODE OF ALARM RECEIVER:

```
package com.huzaihashahbaz.prayeralarmapp;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.media.Ringtone;
import android.media.RingtoneManager;
import android.net.Uri;
import android.widget.Toast;
public class AlarmReceiver extends BroadcastReceiver {
```

```
@Override
public void onReceive(Context context, Intent intent)
{
    Uri alarmUri=
    RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
    if (alarmUri==null)
    {

alarmUri=RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);
    }
    Ringtone ringtone=RingtoneManager.getRingtone(context,alarmUri);
    ringtone.play();
}
}
```

13.5 OUTPUT OF MAIN ACTIVITY:



Figure 13.1:Main Activity of prayer alarm app

CHAPTER#14 Multiple Activities and Intent AND CODES

14.1 Multiple Activities and Intent:

The Prayer Alarm app is a flexible tool for spiritual practice that easily incorporates a number of different activities. It enables users to program personalized alarms for prayer times, with uplifting messages and

subdued reminders to promote regular devotion. Users can easily move through a variety of prayer routines and access materials like scripture readings and meditation prompts thanks to the user-friendly interface. In order to ensure accuracy and inclusivity, the app also provides insights into prayer schedules based on geographic location and religious preferences. In general, it helps people develop a closer spiritual bond and gives them the tools to practice mindfulness and devotion every day.

14.2 JAVA CODE OF MAINACTIVITYINTENT:

```
package intent;
import android.app.AlarmManager;
import android.app.PendingIntent;
import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.TimePicker;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import com.huzaihashahbaz.prayeralarmapp.AlarmReceiver;
import com.huzaihashahbaz.prayeralarmapp.R;
import java.util.Calendar;
public class MainActivity extends AppCompatActivity {
    private TimePicker timePicker;
    private Button buttonSetAlarm;
    private TextView textViewStatus;
    private AlarmManager alarmManager;
    private PendingIntent pendingIntent;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_mainintent);
        timePicker=findViewById(R.id.timePicker);
        buttonSetAlarm=findViewById(R.id.buttonSetAlarm);
        textViewStatus=findViewById(R.id.textViewStatus);
        alarmManager=(AlarmManager)
        getSystemService(Context.ALARM_SERVICE);
```

```

        buttonSetAlarm.setOnClickListener(v -> setAlarm());
    }
    private void setAlarm()
    {
        int hour=timePicker.getCurrentHour();
        int minute=timePicker.getCurrentMinute();
        Calendar calendar=Calendar.getInstance();
        calendar.set(Calendar.HOUR_OF_DAY,hour);
        calendar.set(Calendar.MINUTE,minute);
        calendar.set(Calendar.SECOND,0);
        Intent intent=new Intent(this, AlarmReceiver.class);

        pendingIntent=pendingIntent.getBroadcast(this,0,intent,PendingIntent.FLAG_UP
        DATE_CURRENT);

        alarmManager.setExact(AlarmManager.RTC_WAKEUP,calendar.getTimeInMillis(
        ),pendingIntent);
        String alarmTime=String.format("%02d:%02d",hour,minute);
        textViewStatus.setText("Alarm set for"+alarmTime);
        Toast.makeText(this, "Alarm set for", Toast.LENGTH_SHORT).show();
    }
}

```

14.3 XML CODE OF INTENT:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
    <TimePicker
        android:id="@+id/timePicker"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_gravity="center_horizontal"
        android:layout_marginBottom="20dp"
        android:padding="16dp"
        android:timePickerMode="spinner" />

```

```

<Button
    android:id="@+id/buttonSetAlarm"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Set Alarm"
    android:padding="16dp"/>
<TextView
    android:id="@+id/textViewStatus"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text=""
    android:padding="16dp"/>
</LinearLayout>

```

14.4 JAVA CODE OF ALARM RECEIVER:

```

package intent;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.media.Ringtone;
import android.media.RingtoneManager;
import android.net.Uri;
import android.widget.Toast;
public class AlarmReceiver extends BroadcastReceiver {
    @Override
    public void onReceive(Context context, Intent intent)
    {
        Uri alarmUri=
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
        if (alarmUri==null)
        {

alarmUri=RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);
        }
        Ringtone ringtone=RingtoneManager.getRingtone(context,alarmUri);
        ringtone.play();
    }
}

```

14.5 OUTPUT OF INTENT:

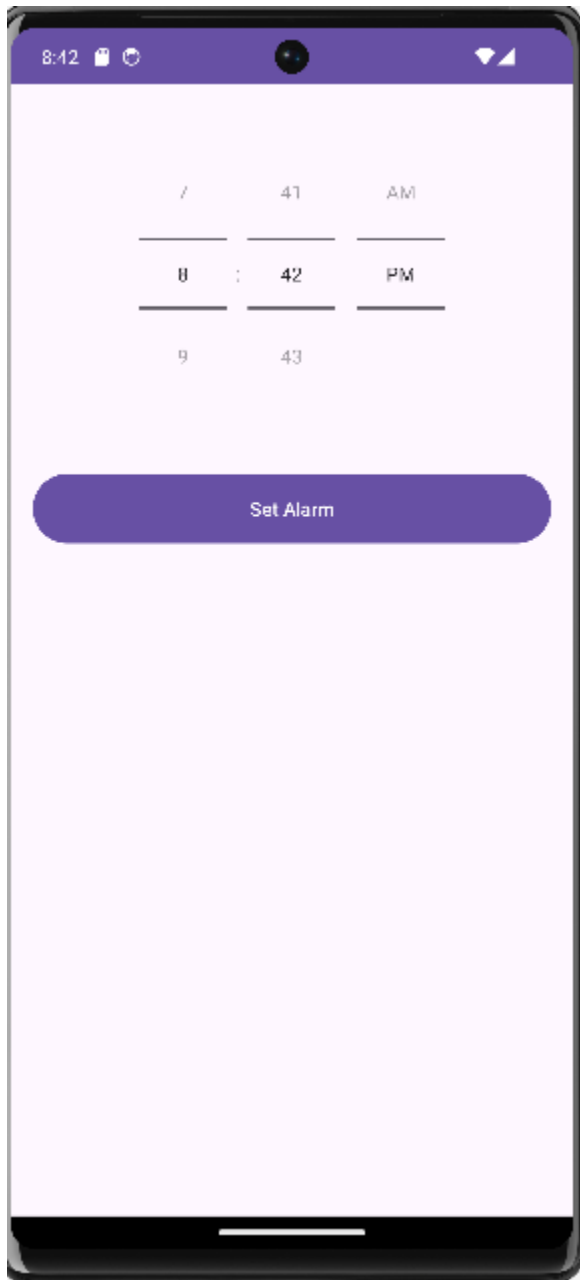


Figure 14.2: Intent activity of prayer alarm app

CHAPTER#15 LIST VIEW AND ARRAY ADAPTER AND CODE

15.1 LIST VIEW AND ARRAY ADAPTER:

A List-View is used by a prayer alarm app to show the times of prayer. An Array-Adapter is used to populate the List-View by binding the data to the List-View. In the List-View, every prayer time is shown as an item.

The data source is managed and adapted to the List-View by the Array-Adapter, which also handles updates dynamically. To improve their prayer experience, users can easily set alarms from the List-View for designated prayer times.

15.2 JAVA CODE OF LIST VIEW AND ARRAY ADAPTER:

```
package ARRAYADAPTERANDLISTVIEW;
import android.app.AlarmManager;
import android.app.PendingIntent;
import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.ListView;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import com.huzaifahshahbaz.prayeralarmapp.AlarmReceiver;
import com.huzaifahshahbaz.prayeralarmapp.R;
import java.util.Calendar;
public class MainActivity extends AppCompatActivity {
    private ListView listViewPrayers;
    private Button buttonSetAlarm;
    private TextView textViewStatus;
    private AlarmManager alarmManager;
    private PendingIntent pendingIntent;
    private String selectedTime;
    private String[] prayerTimes = {"3:20", "12:05", "5:03", "7:11", "8:52"};
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_mainarrayadapterlistview);
        listViewPrayers = findViewById(R.id.listViewPrayers);
        buttonSetAlarm = findViewById(R.id.buttonSetAlarm);
        textViewStatus = findViewById(R.id.textViewStatus);
        alarmManager = (AlarmManager)
getSystemService(Context.ALARM_SERVICE);
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
```

```

android.R.layout.simple_list_item_single_choice, prayerTimes);
    listViewPrayers.setAdapter(adapter);
    listViewPrayers.setChoiceMode(ListView.CHOICE_MODE_SINGLE);
    listViewPrayers.setOnItemClickListener((parent, view, position, id) ->
selectedTime = prayerTimes[position]);
    buttonSetAlarm.setOnClickListener(v -> setAlarm());
}
private void setAlarm() {
    if (selectedTime == null) {
        Toast.makeText(this, "Please select a prayer time",
Toast.LENGTH_SHORT).show();
        return;
    }
    String[] timeParts = selectedTime.split(":");
    int hour = Integer.parseInt(timeParts[0]);
    int minute = Integer.parseInt(timeParts[1]);
    Calendar calendar = Calendar.getInstance();
    calendar.set(Calendar.HOURL_OF_DAY, hour);
    calendar.set(Calendar.MINUTE, minute);
    calendar.set(Calendar.SECOND, 0);
    Intent intent = new Intent(this, AlarmReceiver.class);
    pendingIntent = PendingIntent.getBroadcast(this, 0, intent,
PendingIntent.FLAG_UPDATE_CURRENT);
    alarmManager.setExact(AlarmManager.RTC_WAKEUP,
calendar.getTimeInMillis(), pendingIntent);
    textViewStatus.setText("Alarm set for " + selectedTime);
    Toast.makeText(this, "Alarm set for " + selectedTime,
Toast.LENGTH_SHORT).show();
}
}

```

15.3 XML CODE OF ARRAYADAPTERANDLISTVIEW:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"

```

```

tools:context="ARRAYADAPTERANDLISTVIEW.MainActivity">
<ListView
    android:id="@+id/listViewPrayers"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="16dp"/>
<Button
    android:id="@+id/buttonSetAlarm"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Set Alarm"
    android:padding="16dp"/>
</LinearLayout>

```

15.4 JAVA CODE OF BROADCASTRECEIVER:

```

package ARRAYADAPTERANDLISTVIEW;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.media.Ringtone;
import android.media.RingtoneManager;
import android.net.Uri;

public class AlarmBroadcastReceiver extends BroadcastReceiver {
    @Override
    public void onReceive(Context context, Intent intent)
    {
        Uri alarmUri=
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
        if (alarmUri==null)
        {
alarmUri=RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);
        }
        Ringtone ringtone=RingtoneManager.getRingtone(context,alarmUri);
        ringtone.play();
    }
}

```

15.5 JAVA CODE OF ALARM RECEIVER:

```
package com.huzaifashahbaz.prayeralarmapplistviewandarrayadapter;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.media.Ringtone;
import android.media.RingtoneManager;
import android.net.Uri;
import android.widget.Toast;
public class AlarmReceiver extends BroadcastReceiver {
    @Override
    public void onReceive(Context context, Intent intent)
    {
        Uri alarmUri=
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
        if (alarmUri==null)
        {

alarmUri=RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);
        }
        Ringtone ringtone=RingtoneManager.getRingtone(context,alarmUri);
        ringtone.play();
    }
}
```


15.6 OUTPUT OF LIST VIEW AND ARRAY ADAPTER:

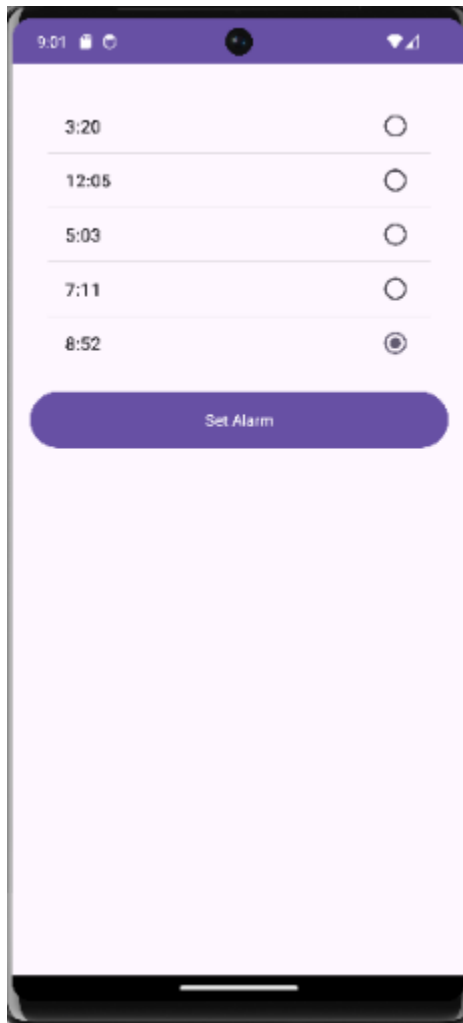


Figure 15.3:ARRAY ADAPTER AND LIST VIEW OF PRAYER ALARM APP

CHAPTER#16 LIST VIEW AND BASE ADAPTER AND CODES

16.1 LIST VIEW AND BASE ADAPTER:

An Android UI element called a List View shows a list of scrollable elements arranged vertically. Developers can specify how data is shown by creating custom adapters for List Views using the Base-Adapter abstract class. A Base-Adapter can be used to alter the appearance and functionality of each item in an alarm clock app, such as the prayer time display and alarm setting capabilities. A List View can present a list of prayer times. This combination makes it easier to manage alarms and prayer routines with an intuitive interface.

16.2 JAVA CODE OF BASEADAPTERANDLISTVIEW:

```
package BASEADAPTERANDLISTVIEW;
import android.annotation.SuppressLint;
import android.app.AlarmManager;
import android.app.PendingIntent;
import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.ListView;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import com.huzafashahbaz.prayeralarmapp.R;
import java.util.Calendar;
public class MainActivity extends AppCompatActivity
{
    private ListView listViewPrayers;
    private Button buttonSetAlarm;
    private TextView textViewStatus;
    private AlarmManager alarmManager;
    private PendingIntent pendingIntent;
    private PrayerTimeAdapter adapter;
    private String[] prayerTimes = {"3:20", "12:05", "5:03", "7:11", "8:52"};
    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        listViewPrayers = findViewById(R.id.listViewPrayers);
        buttonSetAlarm = findViewById(R.id.buttonSetAlarm);
        textViewStatus = findViewById(R.id.textViewStatus);
        alarmManager = (AlarmManager)
getSystemService(Context.ALARM_SERVICE);
        adapter = new PrayerTimeAdapter(this, prayerTimes);
        listViewPrayers.setAdapter(adapter);
        buttonSetAlarm.setOnClickListener(v -> setAlarm());
    }
    private void setAlarm() {
        String selectedTime = adapter.getSelectedTime();
```

```

        if (selectedTime == null) {
            Toast.makeText(this, "Please select a prayer time",
                Toast.LENGTH_SHORT).show();
            return;
        }
        String[] timeParts = selectedTime.split(":");
        int hour = Integer.parseInt(timeParts[0]);
        int minute = Integer.parseInt(timeParts[1]);
        Calendar calendar = Calendar.getInstance();
        calendar.set(Calendar.HOUR_OF_DAY, hour);
        calendar.set(Calendar.MINUTE, minute);
        calendar.set(Calendar.SECOND, 0);
        Intent intent = new Intent(this, AlarmReceiver.class);
        PendingIntent pendingIntent = PendingIntent.getBroadcast(this, 0, intent,
            PendingIntent.FLAG_UPDATE_CURRENT);
        alarmManager.setExact(AlarmManager.RTC_WAKEUP,
            calendar.getTimeInMillis(), pendingIntent);
        textViewStatus.setText("Alarm set for " + selectedTime);
        Toast.makeText(this, "Alarm set for " + selectedTime,
            Toast.LENGTH_SHORT).show();
    }
}

```

16.3 XML CODE OF BASEADAPTERANDLISTVIEW:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
    <ListView
        android:id="@+id/listViewPrayers"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:padding="16dp"/>
    <Button
        android:id="@+id/buttonSetAlarm"

```

```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Set Alarm"
        android:padding="16dp"/>
<TextView
    android:id="@+id/textViewStatus"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text=""
    android:padding="16dp"/>
</LinearLayout>

```

16.4 JAVA CODE OF ALARM RECEIVER:

```

package BASEADAPTERANDLISTVIEW;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.media.Ringtone;
import android.media.RingtoneManager;
import android.net.Uri;
import android.widget.Toast;
public class AlarmReceiver extends BroadcastReceiver {
    @Override
    public void onReceive(Context context, Intent intent)
    {
        Uri alarmUri=
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
        if (alarmUri==null)
        {

alarmUri=RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);
        }
        Ringtone ringtone=RingtoneManager.getRingtone(context,alarmUri);
        ringtone.play();
    }
}

```

16.5 JAVA CODE OF PRAYERTIMEADAPTER:

```
package BASEADAPTERANDLISTVIEW;
import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.RadioButton;
import android.widget.TextView;
import com.huzafashahbaz.prayeralaramapp.R;
public class PrayerTimeAdapter extends BaseAdapter {
    private Context context;
    private String[] prayerTimes;
    private int selectedPosition = -1;
    public PrayerTimeAdapter(Context context, String[] prayerTimes) {
        this.context = context;
        this.prayerTimes = prayerTimes;
    }
    @Override
    public int getCount() {
        return prayerTimes.length;
    }
    @Override
    public Object getItem(int position) {
        return prayerTimes[position];
    }
    @Override
    public long getItemId(int position) {
        return position;
    }
    @Override
    public View getView(int position, View convertView, ViewGroup parent) {
        ViewHolder holder;

        if (convertView == null) {
            convertView =
                LayoutInflater.from(context).inflate(R.layout.activity_prayer_time_item, parent,
                    false);
            holder = new ViewHolder();
            holder.textViewTime = convertView.findViewById(R.id.textViewTime);
```

```

        holder.radioButton = convertView.findViewById(R.id.radioButton);
        convertView.setTag(holder);
    } else {
        holder = (ViewHolder) convertView.getTag();
    }

    holder.textViewTime.setText(prayerTimes[position]);
    holder.radioButton.setChecked(position == selectedPosition);
    convertView.setOnClickListener(v -> {
        selectedPosition = position;
        notifyDataSetChanged();
    });
    return convertView;
}

public String getSelectedTime() {
    if (selectedPosition != -1) {
        return prayerTimes[selectedPosition];
    }
    return null;
}

private static class ViewHolder {
    TextView textViewTime;
    RadioButton radioButton;
}
}

```

16.6 XML CODE OF PRAYERTIMEADAPTER:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="horizontal"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="8dp">
    <TextView
        android:id="@+id/textViewTime"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"

```

```
        android:textSize="18sp"/>
<RadioButton
    android:id="@+id/radioButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
</LinearLayout>
```

16.7 OUTPUT OF BASEADAPTERANDLISTVIEW:

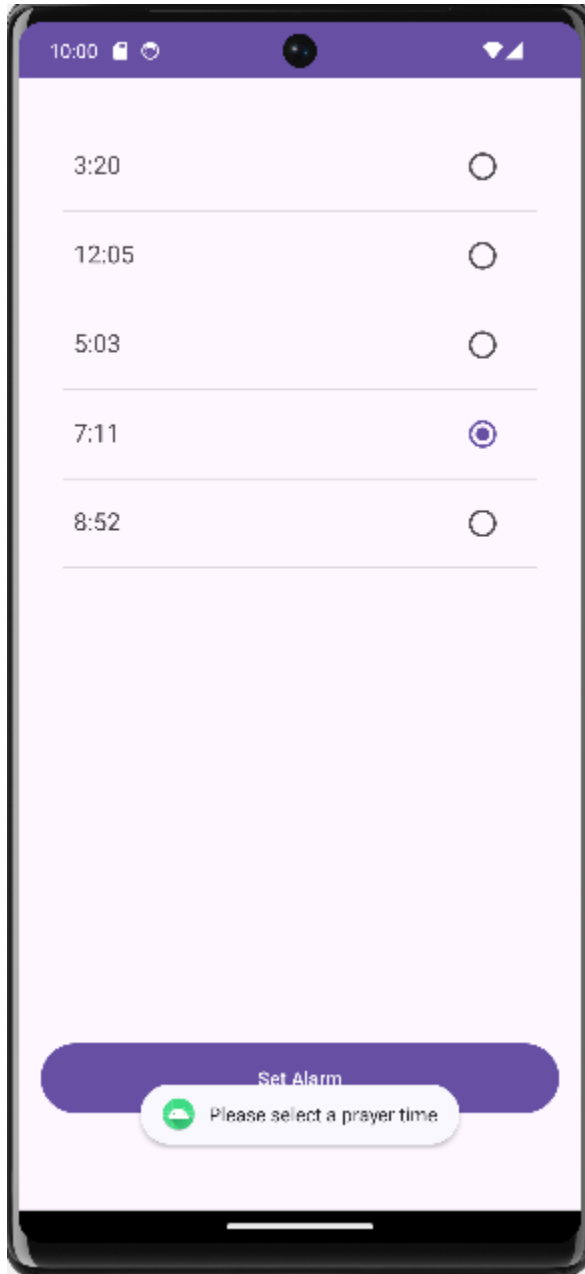


Figure 16.4:BASE ADAPTER AND LISTVIEW OF PRAYER ALARM

CHAPTER#17 SHARED PREFERENCES

17.1 SHARED PREFERENCES:

In Android, key-value pairs are persistently stored in Shared Preferences. You can utilize Shared choices to store settings for prayer times and alarm choices for prayer alarm apps. This lets users personalize their alarms and guarantees that their selections stay in effect even after the app is closed. The software can effectively handle alarm settings by leveraging Shared Preferences, giving users a smooth experience while they participate in their prayer routines.

17.2 JAVA CODE OF SHARED PREFERENCES:

```
package com.huzaifashahbaz.prayeralarmsharedpreferences;
import android.app.AlarmManager;
import android.app.PendingIntent;
import android.content.Context;
import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.TimePicker;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;
public class MainActivity extends AppCompatActivity {
    private TimePicker timePicker;
    private Button setAlarmButton;
    private TextView prayerTimeTextView;
    private static final String PREFS_NAME = "PrayerAlarmPrefs";
    private static final String PREF_HOUR = "hour";
    private static final String PREF_MINUTE = "minute";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        timePicker = findViewById(R.id.timePicker);
        setAlarmButton = findViewById(R.id.setAlarmButton);
        prayerTimeTextView = findViewById(R.id.prayerTimeTextView);
        SharedPreferences preferences = getSharedPreferences(PREFS_NAME,
```



```

MODE_PRIVATE);
    int hour = preferences.getInt(PREF_HOUR, 0);
    int minute = preferences.getInt(PREF_MINUTE, 0);
    updatePrayerTimeTextView(hour, minute);
    setAlarmButton.setOnClickListener(v -> {
        int selectedHour = timePicker.getHour();
        int selectedMinute = timePicker.getMinute();
        setAlarm(selectedHour, selectedMinute);
        savePrayerTime(selectedHour, selectedMinute);
        updatePrayerTimeTextView(selectedHour, selectedMinute);
    });
}

private void setAlarm(int hour, int minute) {
    AlarmManager alarmManager = (AlarmManager)
getSystemService(Context.ALARM_SERVICE);
    Intent intent = new Intent(this, AlarmReceiver.class);
    PendingIntent pendingIntent = PendingIntent.getBroadcast(this, 0, intent, 0);
    Calendar calendar = Calendar.getInstance();
    calendar.set(Calendar.HOUR_OF_DAY, hour);
    calendar.set(Calendar.MINUTE, minute);
    calendar.set(Calendar.SECOND, 0);
    alarmManager.setExact(AlarmManager.RTC_WAKEUP,
calendar.getTimeInMillis(), pendingIntent);
    Toast.makeText(this, "Alarm set for " + hour + ":" + minute,
Toast.LENGTH_SHORT).show();
}

private void savePrayerTime(int hour, int minute) {
    SharedPreferences.Editor editor = getSharedPreferences(PREFS_NAME,
MODE_PRIVATE).edit();
    editor.putInt(PREF_HOUR, hour);
    editor.putInt(PREF_MINUTE, minute);
    editor.apply();
}

private void updatePrayerTimeTextView(int hour, int minute) {
    String timeText = "Prayer Time: " + String.format("%02d:%02d", hour,
minute);
    prayerTimeTextView.setText(timeText);
}
}

```

17.3 XML CODE OF SHARED PREFERENCES:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
    <TextView
        android:id="@+id/prayerTimeTextView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Prayer Time"
        android:textSize="18sp"
        android:layout_marginBottom="16dp" />
    <TimePicker
        android:id="@+id/timePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:timePickerMode="spinner" />
    <Button
        android:id="@+id/setAlarmButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Set Alarm"
        android:layout_marginTop="16dp" />
</LinearLayout>
```

17.4 JAVA CODE OF ALARM RECEIVER:

```
package com.huzaifashahbaz.prayeralarmsharedpreferences;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.media.Ringtone;
import android.media.RingtoneManager;
import android.net.Uri;
import android.widget.Toast;
public class AlarmReceiver extends BroadcastReceiver {
    @Override
```

```
public void onReceive(Context context, Intent intent)
{
    Uri alarmUri=
    RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
    if (alarmUri==null)
    {

alarmUri=RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);
    }
    Ringtone ringtone=RingtoneManager.getRingtone(context,alarmUri);
    ringtone.play();
    }
}
```

17.5 OUTPUT OF SHARED PREFERENCES:

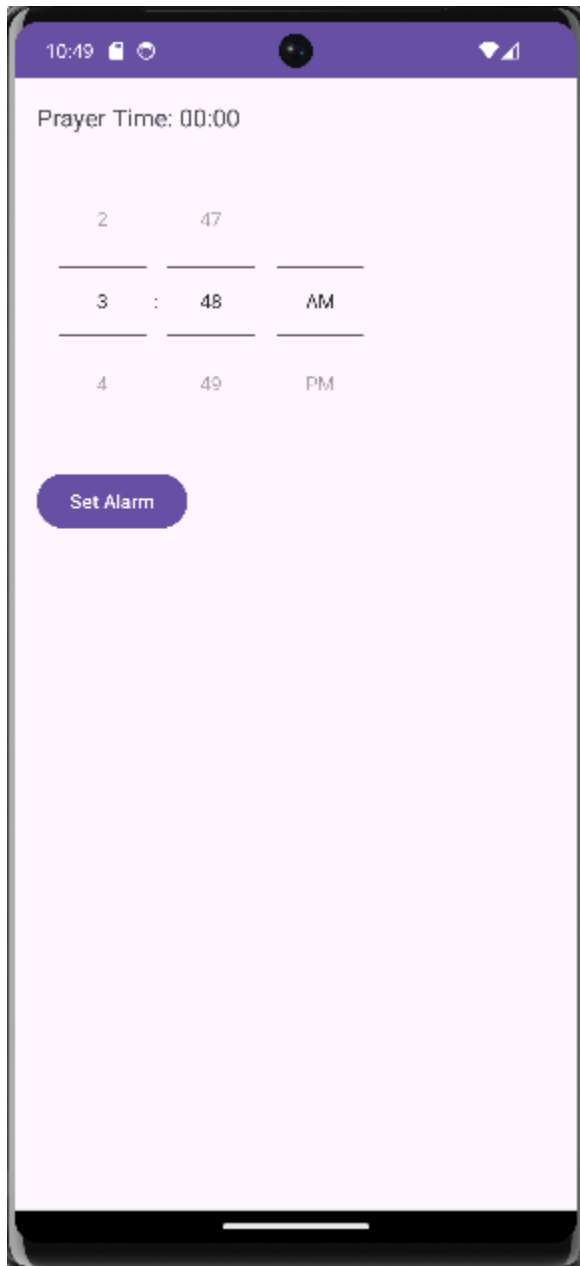


Figure 17.5:SHARED PERFERNCES OF PRAYER ALARM

CHAPTER#18 VIEW PAGER AND CODES

18.1 VIEW PAGER:

The view pager of prayer alarm app following are:

18.1.1 Welcome Screen:

An introduction to the software with a brief summary and a "Get Started" button.

18.1.2 Set Prayer Times:

The user picks their location to set precise prayer times or manually enters prayer times.

18.1.3 Alarm Settings:

The user can set alarm tones, vibration patterns, and pre-prayer reminders for each prayer time.

18.1.4 Qibla Direction:

An interactive compass screen that displays the Qibla direction according to the user's location

18.1.5 Daily Prayer Tracker:

A screen that displays a calendar or checklist for monitoring daily prayers, as well as choices to mark prayers as accomplished.

18.2 JAVA CODE OF VIEW PAGER:

```
package com.huzaifashahbaz.prayeralarmviewpager;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.viewpager2.widget.ViewPager2;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        ViewPager2 viewPager = findViewById(R.id.viewPager);
        ViewPagerAdapter adapter = new ViewPagerAdapter(this);
        viewPager.setAdapter(adapter);
    }
}
```

18.3 XML CODE OF VIEW PAGER:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
```

```

tools:context=".MainActivity">
<androidx.viewpager2.widget.ViewPager2
    android:id="@+id/viewPager"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
</LinearLayout>

```

18.4 JAVA CODE OF PRAYER FRAGMENT:

```

package com.huzaihashahbaz.prayeralarmviewpager;
import android.os.Bundle;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
public class PrayerFragment extends Fragment {
    private static final String ARG_TEXT = "argText";
    public static PrayerFragment newInstance(String text) {
        PrayerFragment fragment = new PrayerFragment();
        Bundle args = new Bundle();
        args.putString(ARG_TEXT, text);
        fragment.setArguments(args);
        return fragment;
    }
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable
ViewGroup container, @Nullable Bundle savedInstanceState) {
        return inflater.inflate(R.layout.fragment_prayer, container, false);
    }
    @Override
    public void onViewCreated(@NonNull View view, @Nullable Bundle
savedInstanceState) {
        super.onViewCreated(view, savedInstanceState);
        TextView textView = view.findViewById(R.id.textView);
        if (getArguments() != null) {
            textView.setText(getArguments().getString(ARG_TEXT));

```

```

    }
}

```

18.5 XML CODE OF PRAYER FRAGMENT:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center">
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Prayer Time"
        android:textSize="24sp"/>
</LinearLayout>

```

18.6 JAVA CODE OF VIEWPAGERADAPTER:

```

package com.huzaifashahbaz.prayeralarmviewpager;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentActivity;
import androidx.viewpager2.adapter.FragmentStateAdapter;
public class ViewPagerAdapter extends FragmentStateAdapter {
    private static final int NUM_PAGES = 5;
    public ViewPagerAdapter(@NonNull FragmentActivity fragmentActivity) {
        super(fragmentActivity);
    }
    @NonNull
    @Override
    public Fragment createFragment(int position) {
        return PrayerFragment.newInstance("Prayer Time " + (position + 1));
    }
    @Override
    public int getItemCount() {
        // Return the number of pages
    }
}

```

```
        return NUM_PAGES;  
    }  
}
```

18.7 OUTPUT OF VIEW PAGER:

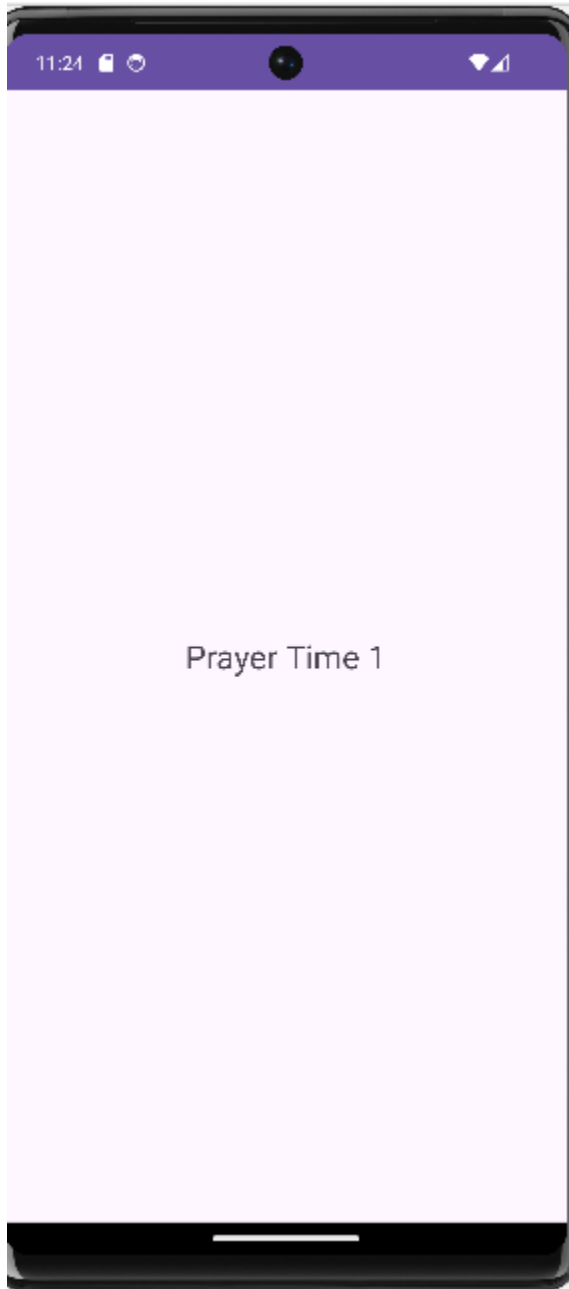


Figure18.6:VIEW PAGER OF PRAYER ALARM APP

Chapter#19 Conclusion:

19.1 Conclusion:

In conclusion, this project aims to create an Android prayer alarm application with features like alarm setting, customization, time-zone adjustment, and calendar integration. It offers a user-friendly interface and efficient code implementation for managing daily prayer routines.